DATA ANALYTICS PROJECT REPORT

NASSAU CANDY DISTRIBUTOR

Product Line Profitability & Margin Performance Analysis

Exploratory Data Analysis, Profitability Diagnostics, Profit Concentration & Interactive Dashboard

Transactions	Products	Divisions	Overall Gross Margin
10,194	**15**	**3**	**65.91%**

Prepared as part of a Data Analytics Internship Project

Tools: Python • Pandas • Streamlit • Excel / CSV

Research Paper | EDA • Insights • Recommendations

Submission Report

2026

. Executive Overview

This research paper presents an analytical assessment of Nassau Candy Distributor's sales, cost, gross profit and gross-margin performance. The analysis covers 10,194 transaction records, 15 products, 3 divisions and 4 regions. The business generated $141,783.63 in total sales and $93,442.80 in gross profit, resulting in an overall gross margin of 65.91%.

A major finding is the high concentration of profitability: the top five products account for 95.06% of total gross profit. This indicates that protecting the availability, pricing and margin performance of these products should be a priority.

2. Project Objectives

- Measure overall sales, cost, units and gross-profit performance.
- Identify the products contributing most to gross profit and margin.
- Compare profitability across divisions and regions.
- Diagnose the relationship between cost and sales.
- Identify low-margin products that may require attention.
- Assess profit concentration using Pareto-style and dependency indicators.
- Build an interactive Streamlit dashboard for management-oriented exploration.
- Translate analytical findings into practical business recommendations.

3. Dataset Description

The analysis uses the Nassau Candy Distributor transactional dataset. It contains 10,194 rows and includes order, customer, geographic, product and financial fields. Key financial variables are Sales, Cost, Gross Profit and Units.

Dataset Metric	Value
Transaction records	10,194
Unique orders	8,549
Unique customers	5,044
Unique products	15
Divisions	3
Regions	4
Total units	38,654

4. Methodology

The analysis was performed using Python and Pandas for data preparation and aggregation, followed by Streamlit for interactive dashboard development. Product-level and division-level summaries were created using grouped aggregations. Gross Margin % was calculated from Gross Profit divided by Sales. Profit contribution and cumulative contribution were used for concentration analysis.

- Loaded the transactional dataset into Pandas.
- Reviewed and prepared relevant date, product, division and financial fields.
- Aggregated Sales, Cost, Gross Profit and Units at product and division levels.
- Calculated Gross Margin % for profitability comparison.
- Created cost-versus-sales diagnostics and margin risk flags.

- Created profit concentration and dependency indicators.
- Added interactive dashboard controls for date, division, margin threshold and product selection.

5. Overall Financial Performance

Metric	Result
Total Sales	$141,783.63
Total Cost	$48,340.83
Total Gross Profit	$93,442.80
Gross Margin	65.91%
Units Sold	38,654

The dataset generated $141,783.63 in sales against $48,340.83 in cost, producing $93,442.80 of gross profit. The resulting gross margin is 65.91%, indicating that a substantial share of sales is retained as gross profit before other operating expenses.

6. Product Profitability Analysis

Product-level analysis was used to identify the strongest contributors to gross profit and to compare margin performance. Products were ranked by gross profit and gross margin percentage.

Product	Sales	Gross Profit	Gross Margin
Wonka Bar - Scrumdiddlyumptious	$27,874.80	$19,357.50	69.44%
Wonka Bar - Triple Dazzle Caramel	$28,485.00	$18,610.20	65.33%
Wonka Bar - Milk Chocolate	$26,867.75	$17,443.37	64.92%
Wonka Bar - Nutty Crunch Surprise	$23,574.95	$16,819.95	71.35%
Wonka Bar - Fudge Mallows	$24,890.40	$16,593.60	66.67%
Lickable Wallpaper	$7,860.00	$3,930.00	50.00%
Wonka Gum	$597.50	$310.70	52.00%
Everlasting Gobstopper	$130.00	$104.00	80.00%
Kazookles	$1,205.75	$92.75	7.69%
Hair Toffee	$76.50	$59.50	77.78%

The top five products generate 95.06% of total gross profit. This concentration creates a clear dependency on a relatively small group of products and makes availability, pricing and margin management for these products particularly important.

7. Division Performance

Division	Sales	Gross Profit	Gross Margin
Chocolate	$131,692.90	$88,824.62	67.45%
Other	$9,663.25	$4,333.45	44.84%
Sugar	$427.48	$284.73	66.61%

The highest gross-profit division in the dataset is Chocolate, contributing $88,824.62 in gross profit with a gross margin of 67.45%. Division-level comparison helps management identify where revenue and profitability are strongest.

8. Regional Performance

Region	Sales	Gross Profit	Gross Margin
Pacific	$46,301.53	$30,485.94	65.84%
Atlantic	$41,197.24	$26,973.70	65.47%
Interior	$32,037.60	$21,282.49	66.43%
Gulf	$22,247.26	$14,700.67	66.08%

Pacific is the leading region by gross profit in the analyzed dataset, with $30,485.94 of gross profit. Regional results can support inventory, sales and market-prioritization decisions.

9. Cost Structure Diagnostics

The cost-versus-sales diagnostic examines whether products with higher cost are also generating higher sales. At transaction level, the correlation between Cost and Sales is 0.959. A positive relationship indicates that higher-cost activity is generally associated with higher sales, but the relationship should be interpreted together with gross margin because sales volume alone does not guarantee strong profitability.

Using a 20% gross-margin threshold as a diagnostic flag, 1 products fall below the threshold. These products should be reviewed for pricing, procurement cost, discounting, product mix and demand efficiency.

10. Profit Concentration & Pareto Analysis

Profit concentration is pronounced: the top five products contribute 95.06% of total gross profit. A Pareto-style ranking orders products from highest to lowest gross profit and tracks cumulative profit contribution. This helps identify how quickly total profitability accumulates across the product portfolio.

Rank	Product	Share of Total Gross Profit
1	Wonka Bar -Scrumdiddlyumptious	20.72%
2	Wonka Bar - Triple Dazzle Caramel	19.92%
3	Wonka Bar - Milk Chocolate	18.67%
4	Wonka Bar - Nutty Crunch Surprise	18.00%
5	Wonka Bar - Fudge Mallows	17.76%

The dependency indicator used in the dashboard measures the share of total gross profit generated by the top products. A high dependency percentage suggests that disruption, stock-outs or margin deterioration in leading products could have a disproportionate effect on total profitability.

11. Key Insights

- The business generated $93,442.80 in gross profit from $141,783.63 in sales.
- Overall gross margin is 65.91%.
- Profitability is highly concentrated: the top five products contribute 95.06% of total gross profit.
- Chocolate is the leading division by gross profit.
- Pacific is the leading region by gross profit.
- 1 products fall below the 20% diagnostic gross-margin threshold.

12. Business Recommendations

- Protect availability of high-profit products through appropriate inventory and replenishment planning.

- Monitor pricing and discount decisions for the top profit-contributing products to avoid unnecessary margin erosion.
- Investigate low-margin products for procurement-cost reduction, pricing adjustments, bundle opportunities or portfolio rationalization.
- Use division and regional profitability metrics alongside sales when setting commercial priorities.
- Track gross margin regularly rather than relying only on sales volume to evaluate performance.
- Use the interactive dashboard as a recurring management-monitoring tool and refresh it with updated transactional data.
- Reduce dependency risk by identifying opportunities to grow profitable secondary products without weakening the core portfolio.

13. Streamlit Dashboard

An interactive Streamlit dashboard was developed as part of the project. It provides product profitability views, division performance comparisons, cost-versus-sales diagnostics, margin risk flags, profit concentration analysis and user controls for date range, division, minimum margin and product selection.

14. Tools & Technologies

Tool / Technology	Purpose
Python	Data analysis and application logic
Pandas	Data loading, transformation and aggregation
Excel / CSV	Source data format
Streamlit	Interactive dashboard development

15. Conclusion

The analysis demonstrates that Nassau Candy Distributor's profitability is driven disproportionately by a small number of products. While the overall gross margin is strong, the concentration of profit creates a dependency risk that should be actively monitored. Combining product-level margin diagnostics, division and regional comparisons, cost-versus-sales analysis and profit concentration indicators provides a practical framework for improving profitability and supporting data-driven management decisions.

Appendix: Submission Deliverables

- Research Paper – EDA, insights and recommendations.
- Streamlit Interactive Dashboard.
- Executive Summary for stakeholder communication.